Hyperspectral Change Detection Using Semi-Supervised Graph Neural Network and Convex Deep Learning

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***Abstract*— With recent advances in hyperspectral remote sensing, hyperspectral change detection (HCD) methods have been developed for precision agriculture and land cover/use monitoring. Among the hyperspectral techniques, those based on convex optimization (CO) and deep learning (DE) have gained particular attention. However, DE often relies on a vast amount of training data (big data) and time-consuming manual labeling tasks in order to learn the inherent patterns of hyperspectral images (HSIs). On the other hand, CO typically requires sophisticated mathematical regularization terms for effectively and adaptively addressing the ill-posed HCD inverse problem. Considering these challenges, we employ the convex deep (CODE) small-data learning theory recently invented for hyperspectral satellite remote sensing and propose a semi-supervised graph neural network to achieve very low labeling rates. Furthermore, the proposed CODE-HCD method exploits hyperspectral data affine geometry to design a convex Q-quadratic norm regularizer, whose mathematical form is very simple and, thereby, ensures a computationally efficient algorithm. The superiority of CODE-HCD over benchmark methods will be demonstrated on several real-world hyperspectral datasets.**

***Index Terms*— Convex optimization (CO), deep learning (DE), hyperspectral change detection (HCD), semi-supervised learning.**

NOMENCLATURE

$\mathbb{R}^n$	n-dimensional Euclidean space.
$\mathbb{R}^{m\times n}$	Set of all the $m \times n$ real-valued matrices.
$\|\cdot\|_F$	Frobenius norm.
$\|\cdot\|_p$	p-norm.
$\text{vec}(\boldsymbol{M})$	Vectorization of matrix $\boldsymbol{M}$.
$[\boldsymbol{v}]_i$	ith entry of vector $\boldsymbol{v}$.
$[\boldsymbol{M}]_{(i,j)}$	(i, j)th entry of matrix $\boldsymbol{M}$.
$\text{Diag}(\boldsymbol{v})$	Diagonal matrix whose ith diagonal entry is $[\boldsymbol{v}]_i$.
$\text{diag}(\boldsymbol{M})$	Column vector whose ith entry is $[\boldsymbol{M}]_{(i,i)}$.
conv	Convex hull [18].
$\mathcal{I}_Z$	$\{1, \ldots, Z\}$ for any given positive integer Z.
$\mathbf{1}$	All-one vector with dimension specified in the subscript.
$\odot$	Hadamard product.

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I. INTRODUCTION

CHANGE detection (CD) plays an essential role in managing disasters [1] and land cover/use monitoring [2], [3], especially under the rapid development of remote sensing technologies. Traditional CD approaches are mainly designed for detecting changed areas through synthetic aperture radar (SAR) and multispectral images (MSIs). Compared to conventional SAR images and MSIs, hyperspectral images (HSIs), which have drawn increasing attention recently, can provide more comprehensive spectral information. Typically, HSIs contain hundreds of spectral bands ranging from the infrared spectrum to the ultraviolet spectrum with high spectral resolution and have promoted profoundly impactful remote sensing applications [4] thanks to the detailed spectral features. In this work, we focus on developing hyperspectral CD (HCD) techniques.

Conventional CD methods typically determine the changed areas relying on various analyses of the multitemporal images. For example, arithmetic-based approaches measure the similarity of the multitemporal images, followed by detecting the changed pixels by setting a threshold. As one of the representative methods, change vector analysis (CVA) [5] determines the similarity by calculating the Euclidean distance between the change vectors to define the changed area. In [6], an improved CVA, termed compressed CVA (C^2VA), compresses the multispectral feature space of change vectors into a 2-D space. Arithmetic-based approaches have demonstrated reasonable performance but often suffer from noise disruption. Transformation-based category transforms the multitemporal images into specific feature spaces to suppress the unchanged pixels and highlight the changed pixels [7], [8]. Iteratively reweighted multivariate alteration detection (IR-MAD) [9] is an extension of multivariate alteration detection (MAD) [10]. To apply CD on HSI data, IR-MAD introduces the application of regularization to prevent detecting uninteresting changes owing to noise and spurious differences. A transformation-based approach that extracts features from non-overlapping blocks of multitemporal images

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by applying principal component analysis (PCA), followed by clustering the feature vectors using k-means [11], called the PCA algorithm [12], should also be mentioned. In slow feature analysis (SFA) [7], the most invariant component is extracted from the bitemporal images, and in the feature space, the changed areas are detected. Transformation-based methods successfully reduce the data dimension and prevent noise disruption, but they do not fully exploit the information contained in the multitemporal images.

Classification-based methods determine the changed areas by applying classifiers in the algorithm. A framework combining support vector machines (SVMs) has been proposed in [13] to enhance detection performance. However, the effectiveness of classification-based methods highly depends on the classifier. Apart from the approaches previously mentioned, advanced models have been adopted on HSIs to further utilize the information embedded in the observed images with explainable physical meaning. For example, the spectral mixture model is introduced in [14], where the abundance maps of the multitemporal images are generated through sparse unmixing with dictionary pruning, followed by detecting changed pixels by calculating the difference between the two abundance maps. Another CD framework, known as tensor decomposition and reconstruction detector (TDRD) [15], exploits the spatial and spectral information through the adoption of a tensor model containing Tucker decomposition and high-order principal component extraction. These methods have considered both spatial and spectral information but utilize just shallow features.

In recent years, some optimization-based methods have been studied to better describe the characteristics of the target change map. CD-ADMM [16] formulates CD as a convex form and solves the problem utilizing a convex optimization (CO) solver called alternating direction method of multipliers (ADMM) [17], [18] with a convergence guarantee. The first regularization term in CD-ADMM limits the area of the changed pixels based on the observation that the changed regions are relatively small, while the second regularizer promotes piecewise smoothness of the change map by introducing anisotropic total variation (TV) regularization [19]. CD-ADMM has successfully detected changed areas using multispectral satellite data if the parameters in the convex criterion have been well-tuned. In addition, regularizers are required to improve the fitting of the problem in CO-based algorithms and often induce mathematically heavy derivations of the closed-form solutions for implementing the convex criterion, which is not friendly for most software developers to adopt CO-related theory.

Recently, deep learning (DE) has achieved state-of-the-art performance in various remote sensing tasks, such as HSI super-resolution [20] and hyperspectral compressed sensing on miniaturized satellite [21]. Also, DE has drawn attention in the CD field and has been adopted to achieve automatic CD algorithms [22]. We have to mention the method of [23], a single-stream framework that classifies the changed areas, with the input being the concatenate of the bitemporal images. In [24], a general end-to-end 2-D convolutional neural network (CNN), called GETNET, proposes a simple mixed-affinity matrix to mine the cross-channel gradient information, followed by using CNN to detect the changed areas. Moreover, given the nature of the input data, which is bitemporal images, two-stream frameworks are commonly designed for solving CD problems [25]. For example, a Siamese structure network, called bilinear CNN (BCNN) [26], considers the relationships between the bitemporal images by combining the output feature maps of a symmetric CNN using a linear outer product operation. Compared to typical CNNs, recurrent neural network (RNN) methods utilize a recurrent hidden state to perform operations on a sequence of vectors or values to consider the temporal information of the multitemporal images [22], [27]. The recurrent 3-D fully convolutional network (Re3FCN) [28], which consists of 3-D convolutions and convolutional long short-term memory (ConvLSTM) [29] units, effectively extracts the spectral spatial–temporal features from the bitemporal HSIs while maintaining the spatial information. A recurrent CNN (ReCNN) is proposed in [30], combing CNN and RNN to extract information on spectral, spatial, and temporal aspects. By designing the deep Siamese CNN (DSCNN), deep Siamese convolutional multiple-layer RNN (SiamCRNN) [31] can extract spatial–spectral features from multitemporal images. Then, delve into the change information between homogeneous or heterogeneous features by multiple-layer RNN (MRNN) constructed by LSTM units. Furthermore, generative adversarial networks (GANs) are recently introduced to solve the challenging CD tasks [27]. A fully connected neural network structure based on GAN called FCD-GAN is proposed in [32]. FCD-GAN has successfully achieved unsupervised, fully supervised, weakly supervised, and newly defined regional supervised tasks. Aside from the above network structure, the attention mechanism, widely used in various computer vision tasks in recent years, enables neural networks to focus on specific regions or features of input data to selectively process and allocate resources to the most relevant parts [33]. In [34], a novel spectral and spatial attention network (S^2AN) for HCD is proposed with a semi-supervised training strategy to handle the insufficient labeled data issue. The designed spectral and spatial attention module aims to calculate to compute spectral/spatial attention scores of the feature maps. These scores are subsequently utilized to weigh the channels and pixels within the feature maps, respectively, to enhance CD-relevant information. Though CNN-based CD methods have achieved good performance, CNN is designed for Euclidean data, and the fixed kernel size inside the network model only allows CNN to learn the local spatial features. In light of this, the graph neural network (GNN) [35] has been proposed as a way of applying DE to non-Euclidean data, allowing them to learn the structure and node information through an end-to-end neural network. GNN has been adopted on HSI processing tasks [36], [37] to better exploit global spatial information using graphs generated by adopting superpixel segmentation on HSIs. GNNs have also been adopted to address the challenging HCD problem. An unsupervised structural relationship graph representation learning framework for multimodal CD (SRGRL-CD) is proposed in [38]. This framework comprises a structural relationship graph convolutional autoencoder (SR-GCAE) that is

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specifically designed to learn the structural information present in both the edges and vertices of the graph. Utilizing deep-level features produced by SR-GCAE, the local and nonlocal structural relationships can be measured, and an adaptive fusion is performed to accentuate the changed pixels while simultaneously suppressing the unchanged pixels. In dual-branch difference amplification GCN (D^2AGCN) [39], a dual-branch network consisting of multilayer graph convolution networks (GCNs) is proposed to fully extract the features embedded in the multitemporal HSIs. In addition, a difference amplification module (DAM) is designed to highlight the different regions and suppress the unchanged areas of the input HSIs, leading to a superior CD performance. Another noticeable GNN-based CD algorithm is the dual-branch framework based on temporal–spatial joint graph attention with complement strategy (CSDBF) [40]. CSDBF extracts the feature from the pixel level and the subpixel level by a dual-branch feature extraction framework, allowing the GNN network to fully exploit both the subtle and large-scale changes information. The concept of the graph attention network (GAT) [41] is also introduced in the temporal–spatial joint GAT module (TSJGAT) to learn temporal–spatial joint correlations from a temporal–spatial affinity matrix. Experiments have shown that DE-based CD methods are superior, and GNN-based approaches are capable of handling limited situations. However, when the labeling/sampling rate is extremely low, as is often the case in practical situations, especially when the acquisition and labeling of training samples from real data are time-consuming and expensive, those DE-based methods (particularly CNN-based methods) can suffer from overfitting and poor performance.

To address the aforementioned issues regarding math-heavy regularization in CO and the requirement of large training samples in DE, this article adopts a recent framework called convex deep (CODE) [42], which blends the strengths of CO and DE to achieve small-data learning with a CODE regularization function of very simple mathematical form. The CODE theory [42] reveals the fact that a rough solution, generated by a simple neural network trained using just small data, could still contain useful information, and such information can be extracted using the mathematically very simple convex $\boldsymbol{Q}$-quadratic function. In this context, the CODE framework consists of a data-fitting term and a simple convex feature extractor, and is proven to be effective in solving highly challenging hyperspectral inverse problems even without labeled big data. The data-fitting term preserves the data fidelity, while the simple convex feature extractor utilizes the information embedded in the rough solution to replace the traditional math-heavy regularization schemes, leading to computationally efficient inverse imaging software [42]. CODE has tackled various hyperspectral inverse problems, such as hyperspectral inpainting [42], hyperspectral signature reconstruction [43], single HSI super-resolution [44], hyperspectral dehazing [45], and satellite image classification [46]. For the first time, this promising CODE theoretical framework is applied in this article to solve the highly challenging HCD problem. To meet practical considerations, this article will develop an HCD method (termed CODE-HCD) under a semi-supervised setting, and a critical aim in practice will be to achieve low labeling rates while keeping strong HCD performances.

Instead of the commonly used Euclidean distance in CVA, our CODE-HCD adopts the spectral similarity to fit changed areas, as it is not only less sensitive to noise but can better measure the similarity between pixels at different time instances, thereby fully exploiting the spectral information contained in the acquired HSIs. A CODE regularization term is used to extract useful features from a rough solution yielded by a multi-head GNN that is trained using just small data. The convex formulation is then solved by ADMM with a convergence guarantee, and the closed-form solutions are derived for computational efficiency. The main contributions of this work are summarized as follows.

1) An algorithm called CODE-HCD well blending the advantages of CO and DE is proposed to address the challenging ill-posed inverse HCD problem. CODE-HCD employs a mathematically simple regularization scheme with closed-form solutions for each algorithmic step derived and demonstrates superior HCD performances even with very low sampling/labeling rates.
2) A convex data-fitting term is designed based on the spectral angle mapper (SAM) [47], rather than using the naive Euclidean distance in the previous CO-based algorithms. This design, together with the convex relaxation strategy bringing the optimization domain into a convex set, better exploits the critical spectral information of the HSIs acquired at different time instances.
3) A simple GNN architecture trained using small learning data (i.e., only about 0.1% training data sampling/labeling rate) is proven to be sufficient for the proposed CODE-HCD to yield good detection performance, making it practically applicable in semi-supervised scenarios. CODE-HCD judiciously detects the changed areas by the analysis of the relations among the adjacent superpixels of the bitemporal HSIs and by a detail-refinement module for obtaining the change map at the pixel level. Therefore, a simple deep feature distilling network consisting of GAT and CNN layers extracts both pixel- and superpixel-level features, thereby effectively generating a preliminary solution for the subsequent regularization.
4) A simple CODE $\boldsymbol{Q}$-quadratic feature extractor is designed to fully utilize the useful information embedded in the aforementioned preliminary solution, effectively replacing traditional math-heavy or nondifferentiable regularizers (e.g., TV and self-similarity functions [48]) commonly used in existing hyperspectral remote sensing software. As will be seen, this induces a computationally efficient HCD algorithm and works very well even just with small data.

The remaining parts of this article are organized as follows. In Section II, we recall some related backgrounds, including the CODE theoretical framework and existing CO-based CD criterion design. In Section III, the design of the proposed HCD algorithm is presented, and convergence analysis is

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discussed. In Section IV, results of real hyperspectral data experiments, together with comparisons with benchmark CD methods, are summarized and discussed. Conclusions are drawn in Section V, while mathematical proofs are given in the Appendix.

II. Related Background

In this section, we briefly recall some related backgrounds for readers to smoothly access the development of the CODE-HCD algorithm. First, we review the CODE small-data learning theory [42], originally developed for hyperspectral satellite data processing (see Section II-A). Then, a related CO-based CD criterion [16] is discussed (see Section II-B). Nomenclature provides a summary of the notations used in this article to facilitate a clearer understanding of the proposed method.

A. CODE Framework

CO and DE are two critical theories for solving imaging inverse problems, which have brought numerous impacts on various remote sensing applications, such as hyperspectral compressed sensing [21], [49] and hyperspectral super-resolution [20], [50]. CO has been widely adopted in state-of-the-art remote sensing imaging inverse algorithms by formulating the problem as a convex criterion, followed by introducing CO optimizers (e.g., ADMM) to implement the criterion. The convex criterion can be divided into two parts, i.e., a data-fitting term and a regularization function. The data-fitting term describes the relationship between the desired solution and the observed image to ensure a reasonable algorithm output. However, as solving the data-fitting problem itself is typically ill-posed, the regularization term is needed to guide the optimizer toward the solution satisfying some desired physical characteristics (e.g., sparsity). Overall, a typical criterion for solving inverse problems can be formulated as [42]

$$\boldsymbol{c}^{\star} = \arg\min_{\boldsymbol{c}} \ \mathrm{DF}(\boldsymbol{c}) + \lambda\, \mathrm{REG}(\boldsymbol{c}) \tag{1}$$

where $\boldsymbol{c}^{\star}$ is the target solution, $\mathrm{DF}(\boldsymbol{c})$ is the data-fitting term, and $\mathrm{REG}(\boldsymbol{c})$ is the regularizer with regularization strength controlled by the parameter $\lambda > 0$. Note that, even if the regularizers are designed as convex functions, their mathematical forms are often complicated (e.g., the graph-embedded self-similarity regularizer [48]) in order to achieve better performance or to address more sophisticated inverse problems (e.g., multiresolution super-resolution for multispectral satellite [48]).

As for DE, big data are typically required to enable sophisticated neural networks to learn the deep features within the training set. However, acquiring big data is time-consuming and expensive, especially for remote sensing/imaging data. To address the above issues, the CODE theoretical framework was proposed in [42] by blending the advantages of CO and DE. Lin et al. [42] posit that, even though the output of a neural network trained on small data is just a roughly estimated weak solution, it still contains useful features for the subsequent regularization task. For instance, in [42], the roughly estimated solution $\boldsymbol{c}_{\mathrm{DE}}$ is able to provide structural information for effectively solving the highly challenging hyperspectral inpainting problem. A simple CODE feature extractor gaining desired information from $\boldsymbol{c}_{\mathrm{DE}}$ can then be selected as the regularization term in (1), and CODE judiciously designs such a convex extractor based on the so-called convex $\boldsymbol{Q}$-quadratic norm function whose mathematical form is very simple. To be more specific, given a positive semidefinite matrix $\boldsymbol{Q}$, the $\boldsymbol{Q}$-quadratic norm of the feature vector $\boldsymbol{v}$ is defined by $\|\boldsymbol{v}\|_{\boldsymbol{Q}}^2 := \boldsymbol{v}^T \boldsymbol{Q} \boldsymbol{v}$ [51]. Thus, the regularizer can be naturally formulated as

$$\mathrm{REG}(\boldsymbol{c}) := \|\boldsymbol{c} - \boldsymbol{c}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2 \tag{2}$$

where $\boldsymbol{Q}$ describes the subspace formed by useful feature vectors (e.g., hyperspectral subspace derived from hyperspectral data geometry [42]). In [42], the PSD matrix $\boldsymbol{Q}$ is designed based on PCA. In the CODE framework, a rough solution $\boldsymbol{c}_{\mathrm{DE}}$ is sufficient, so a simple network architecture is good enough even for a complicated inverse problem [42]. As a result, the $\boldsymbol{Q}$-quadratic norm $\|\cdot\|_{\boldsymbol{Q}}$ in (2) can be considered as a simultaneous feature extraction and fusion operator (SFEFO) [42]. This SFEFO extracts relevant information embedded in the small-data solution $\boldsymbol{c}_{\mathrm{DE}}$ and fuses the information into the final solution $\boldsymbol{c}^{\star}$ during the optimization procedure [42].

The CODE framework and plug-and-play (PnP) are fundamentally different. In hyperspectral inverse imaging problems, the regularization term of the objective function is often nonconvex and too complicated to be explicitly defined. PnP methods [52], [53], [54], [55] approximate the proximal operator for the regularizer without a theoretical convergence guarantee by adopting state-of-the-art algorithms with the same physical meaning. On the other hand, the regularization term in the CODE theory is explicitly defined with a simple convex form so that convergence is always guaranteed, resulting in a highly computationally efficient algorithm. Moreover, deep PnP methods may need big data for the training stage to achieve high-efficacy performance [54], while CODE only needs just a simple neural network structure with small data in the proposed method. For more details, please refer to [42, Sec. II.D].

To summarize, there are three major tasks when adopting the CODE framework. First, the data-fitting term $\mathrm{DF}(\boldsymbol{c})$ should be customized for the considered inverse problem. Second, a neural network should be designed and trained using small data, thereby yielding a rough solution $\boldsymbol{c}_{\mathrm{DE}}$. Third, design the PSD matrix $\boldsymbol{Q}$ in (2) to complete the convex criterion design of (1), and solve (1) to obtain the optimal solution $\boldsymbol{c}^{\star}$ of the considered inverse problem. We will design the proposed CODE-HCD algorithm based on the above three steps, as will be detailed in Section III.

B. Existing CO-Based CD Criterion Design

In the previously mentioned CD-ADMM [16], the CD criterion has been formulated as a convex problem and solved using ADMM. It is noteworthy that the CD-ADMM algorithm is particularly developed for identifying changed regions

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using multispectral satellite images. The convex criterion of CD-ADMM is based on the difference image (DI) method [12] and two assumptions. The first assumption is that the changed regions are relatively small, and the second is that the change pattern is piecewise smooth. Given two multispectral images $\boldsymbol{X} = [\boldsymbol{x}_1, \ldots, \boldsymbol{x}_L] \in \mathbb{R}^{M\times L}$ and $\boldsymbol{Y} = [\boldsymbol{y}_1, \ldots, \boldsymbol{y}_L] \in \mathbb{R}^{M\times L}$ acquired at different time instances, where M is the number of spectral bands, and $L = L_1 \times L_2$ is the total number of pixels covering the same $L_1 \times L_2$ spatial area, based on the above assumptions, the CD criterion for computing the changed map $\boldsymbol{C} \in \{0, 1\}^{L_1\times L_2}$ can be derived as

$$\min_{[C]_{(i,j)}\in\{0,1\}} \quad \sum_{\forall(i,j)} \Big\{ \big\| (\boldsymbol{X} - \boldsymbol{Y}) \mathrm{Diag}\big(\mathrm{vec}\big(\mathbf{1}_{L_1}\mathbf{1}_{L_2}^T - \boldsymbol{C}\big)\big) \big\|_F^2 + \lambda \mathbf{1}_{L_1}^T \boldsymbol{C} \mathbf{1}_{L_2} + \eta \mathrm{TV}(\boldsymbol{C}) \Big\} \tag{3}$$

where $\mathrm{TV}(\cdot)$ is the anisotropic TV operator to meet the second assumption, while the first assumption is promoted by the second term of (3); $\lambda > 0$ and $\eta > 0$ are the regularization weights. As for the first term of (3), it is naturally understood based on the DI method; if the data $\boldsymbol{x}_\ell$ and $\boldsymbol{y}_\ell$ (i.e., the ℓth pixels at two time instances) are unchanged, their difference should be small. Therefore, the first term of criterion (3) can be regarded as the data-fitting term.

Although the objective function of (3) is convex, the data-fitting domain is nonconvex due to the binary-valued constraint, which prevents the adoption of the powerful convex solvers. Such a nonconvex constraint can be relaxed as its convex hull, i.e.,

$$\boldsymbol{C} \in \mathrm{conv}\big\{\{0, 1\}^{L_1\times L_2}\big\} = [0, 1]^{L_1\times L_2} \tag{4}$$

and interpreted as the continuous probability domain. Therefore, the criterion can then be compactly reformulated as the following convex problem:

$$\min_{\boldsymbol{c}\in[0,1]^L} \|(\boldsymbol{X} - \boldsymbol{Y})\mathrm{Diag}(\mathbf{1}_L - \boldsymbol{c})\|_F^2 + \lambda \mathbf{1}_L^T \boldsymbol{c} + \eta\, \mathrm{TV}(\boldsymbol{c}) \tag{5}$$

where $\boldsymbol{c} = \mathrm{vec}(\boldsymbol{C})$. As both the objective function and the constraint set are convex now, ADMM can be adopted to solve the convex CD criterion (5). An experiment using a real multispectral satellite image in [16] demonstrates the superiority of CD-ADMM. Though the concepts therein are elegant, the regularizer induced by the first assumption is particularly designed for land cover/use monitoring and may not be suitable for general scenarios. Besides, the split Bregman method [56], [57], adopted in CD-ADMM [16] for handling the non-differentiable TV regularizer induced by the second assumption, is not computationally very efficient, especially for the large-scale HCD problem that involves more spectral bands than the MSI CD problem. In this article, we avoid those assumptions/regularizers using a single unifying $\boldsymbol{Q}$-quadratic regularization term, achieving highly effective convex HCD criterion, as presented in Section III.

III. Proposed HCD Method

Compared to SAR images and MSIs, HSIs directly provide abundant spectral information for strong endmember/material identifiability [58], [59], where endmember is a standard usage to refer to the material hyperspectral signature [60]. In other words, CD can be naturally considered as to compare the two pixels (at the same location, but at different time instances) to see whether they are the same material or not. Thus, hyperspectral data are very suitable for the CD mission. In this article, we propose a semi-supervised HCD method (i.e., CODE-HCD) by blending CO and DE based on the CODE theory [42] in order to efficiently solve the challenging HCD inverse problem with a very low training sampling rate (i.e., labeling rate). To fully utilize the information contained in the bitemporal HSIs, we design a neural network incorporating CNN and GNN to guide the HCD data-fitting task. The overall structure of the proposed CODE-HCD method is illustrated in Fig. 1.

Before presenting the technical details, the CODE-HCD is briefly described hereinafter. First, the data-fitting term $\mathrm{DF}(\boldsymbol{c})$ utilizing the rich spectral information of HSIs is proposed to depict the HCD problem (see Section III-A). Then, a deep feature distilling neural network is designed to learn both superpixel- and pixel-level features using a limited number of training samples, followed by a CODE regularizer that extracts features from the deep solution to aid in the selection of an optimal change map (see Section III-B). Finally, the customized convex HCD criterion is implemented by ADMM with all the closed-form solutions derived for computational efficiency, and the convergence of the proposed CODE-HCD is mathematically guaranteed (see Section III-C).

A. DF Design for HCD Problem

In [16], the data-fitting term given in (3) is designed for MSIs based on the DI method. Nevertheless, determining the similarity of the bitemporal HSIs $\boldsymbol{X}$ and $\boldsymbol{Y}$ by calculating the Euclidean distance between the pixel values of the two images ignores the spectral relationship and is sensitive to spectral variation. In Fig. 2, the spectral curves of an unchanged pixel from the Hermiston dataset obtained at different time points are displayed. Notably, the curves reveal that the pixel's material composition remains unchanged over time, whereas there are substantial differences in the energy of the pixels. Owing to the spectral variation, directly computing the Euclidean distance between pixel values to obtain the similarity map could yield unreliable results, as will be experimentally demonstrated. Due to the ability of HSIs that can identify the material variations through spectral signatures [4], SAM is introduced in this article to better detect the changes by fully utilizing the spectral information. SAM is insensitive to spectral variation because it measures the angle between two vectors without considering the total energy. The SAM of bitemporal pixels $\boldsymbol{x}_\ell$ and $\boldsymbol{y}_\ell$ representing the same area of Earth's surface can be defined as

$$\mathrm{SAM}(\boldsymbol{x}_\ell, \boldsymbol{y}_\ell) = \arccos\left(\frac{\boldsymbol{x}_\ell^T \boldsymbol{y}_\ell}{\|\boldsymbol{x}_\ell\|_2 \cdot \|\boldsymbol{y}_\ell\|_2}\right). \tag{6}$$

One can see that the SAM value of unchanged pixels in the bitemporal HSIs $\boldsymbol{X}$ and $\boldsymbol{Y}$ should be small since the material composition of the two images is highly similar. Therefore, SAM is adopted in the proposed $\mathrm{DF}(\boldsymbol{C})$ instead of directly calculating the Euclidean distance between $\boldsymbol{X}$ and $\boldsymbol{Y}$. The data-

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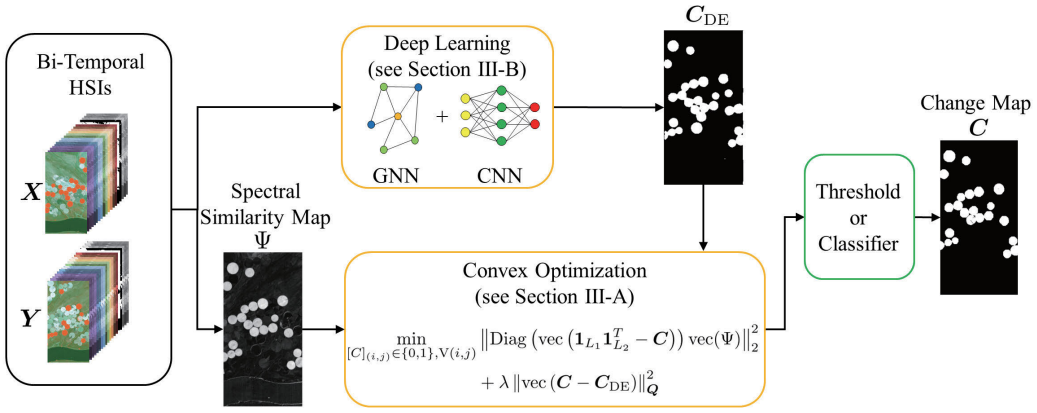


Fig. 1. Overview of the proposed CODE-HCD method that blends the advantages of CO and DE.

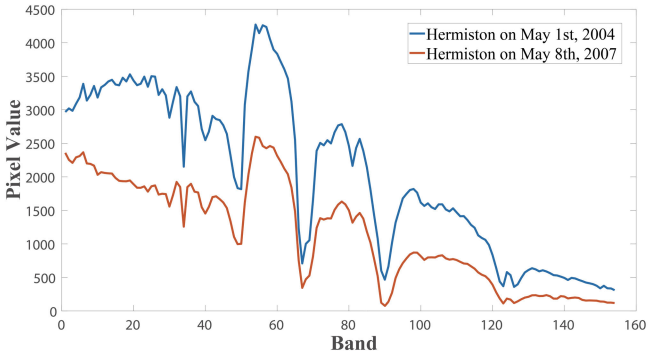


Fig. 2. Hyperspectral curves of an unchanged pixel in the Hermiston dataset acquired at different dates.

fitting term $\mathrm{DF}(\boldsymbol{C})$ can then be formulated as

$$\mathrm{DF}(\boldsymbol{C}) = \left\|\mathrm{Diag}\big(\mathrm{vec}\big(\mathbf{1}_{L_1}\mathbf{1}_{L_2}^T - \boldsymbol{C}\big)\big)\mathrm{vec}(\Psi)\right\|_2^2 \tag{7}$$

where $\Psi \in \mathbb{R}^{L_1\times L_2}$ is the similarity map of $\boldsymbol{X}$ and $\boldsymbol{Y}$ with the ℓth pixel of its vectorization $\mathrm{vec}(\Psi)_\ell = \psi_\ell = \mathrm{SAM}(\boldsymbol{x}_\ell, \boldsymbol{y}_\ell)$ for $\ell \in \mathcal{I}_L$. By combining the proposed $\mathrm{DF}(\boldsymbol{C})$ with the CODE framework given in (1) and (2), we can formulate the HCD problem as

$$\min_{[C]_{(i,j)}\in\{0,1\} \quad \forall(i,j)} \left\|\mathrm{Diag}\big(\mathrm{vec}\big(\mathbf{1}_{L_1}\mathbf{1}_{L_2}^T - \boldsymbol{C}\big)\big)\mathrm{vec}(\Psi)\right\|_2^2 + \lambda\|\mathrm{vec}(\boldsymbol{C} - \boldsymbol{C}_{\mathrm{DE}})\|_{\boldsymbol{Q}}^2 \tag{8}$$

where the deep solution $\boldsymbol{C}_{\mathrm{DE}} \in \{0,1\}^{L_1\times L_2}$ learning deep information from a simple network structure will be detailed in Section III-B. The HCD criterion (8) includes a nonconvex constraint, which precludes the use of powerful convex solvers that often guarantee convergence. By relaxing the constraint set into its convex hull [18], one can compactly reformulate the HCD criterion into a convex form as

$$\min_{\boldsymbol{c}\in[0,1]^L} \|\mathrm{Diag}(\mathbf{1}_L - \boldsymbol{c})\psi\|_2^2 + \lambda\|\boldsymbol{c} - \boldsymbol{c}_{\mathrm{DE}}\|_{\boldsymbol{Q}}^2 \tag{9}$$

where $\boldsymbol{c}_{\mathrm{DE}} = \mathrm{vec}(\boldsymbol{C}_{\mathrm{DE}})$.

Up until now, we have completed the first step to propose an HCD algorithm based on the CODE theory, that is, to customize a convex criterion for the HCD problem. Next, we will need to design a deep regularization term to extract useful features from limited training samples for guiding the data-fitting term $\mathrm{DF}(\boldsymbol{c})$.

B. Graph Attention Change Detection Network Design and DE Regularization Design

As suggested by the CODE small data learning theory [42], we just need to obtain a roughly estimated solution $\boldsymbol{c}_{\mathrm{DE}} = \mathrm{vec}(\boldsymbol{C}_{\mathrm{DE}})$ learned from limited training samples for the subsequent deep regularization in (9). Changed pixels are often observed to form continuous areas rather than being scattered, motivating us to study the changed information at the superpixel level for computational efficiency. Another key observation is that, if a small area (superpixel) is changed, it is likely that the relationship of the superpixel with its neighboring superpixels will also be altered. Therefore, this further motivates us to adopt GNN to study the relationship between adjacent superpixels in the bitemporal HSIs for more effectively detecting the changed areas.

Moreover, compared to the standard GNN, the attention mechanism of GAT [41] specifies different importance levels to the neighboring nodes to focus on digging the local information at the superpixel level. With this in mind, we introduce GAT layers in our network architecture to focus on the interaction between adjacent superpixels. It is important to note that the architecture of the neural network that generates $\boldsymbol{c}_{\mathrm{DE}}$ can be very simple based on the fact that the CODE theory is able to learn useful information from a roughly estimated solution. With this in mind, we propose an unadorned graph attention CD network (GACDN) f_θ, composed of basic GAT/CNN layers, to extract deep features embedded in the small training data. As will be experimentally demonstrated, f_θ provides a reasonable output for the regularization in (9) even under limited training samples. In this section, we describe how to generate a change map from the proposed deep feature distilling network, denoted by $\boldsymbol{C}_{\mathrm{DE}} = f_\theta(\boldsymbol{X}, \boldsymbol{Y})$. The network f_θ consists of three parts, as shown in Fig. 3. First, the bitemporal HSIs $\boldsymbol{X}$ and $\boldsymbol{Y}$ are converted into graph structure by dividing them into superpixels using the superpixel segmentation module. Second, superpixel-level information is extracted by GAT layers learning the relationship between

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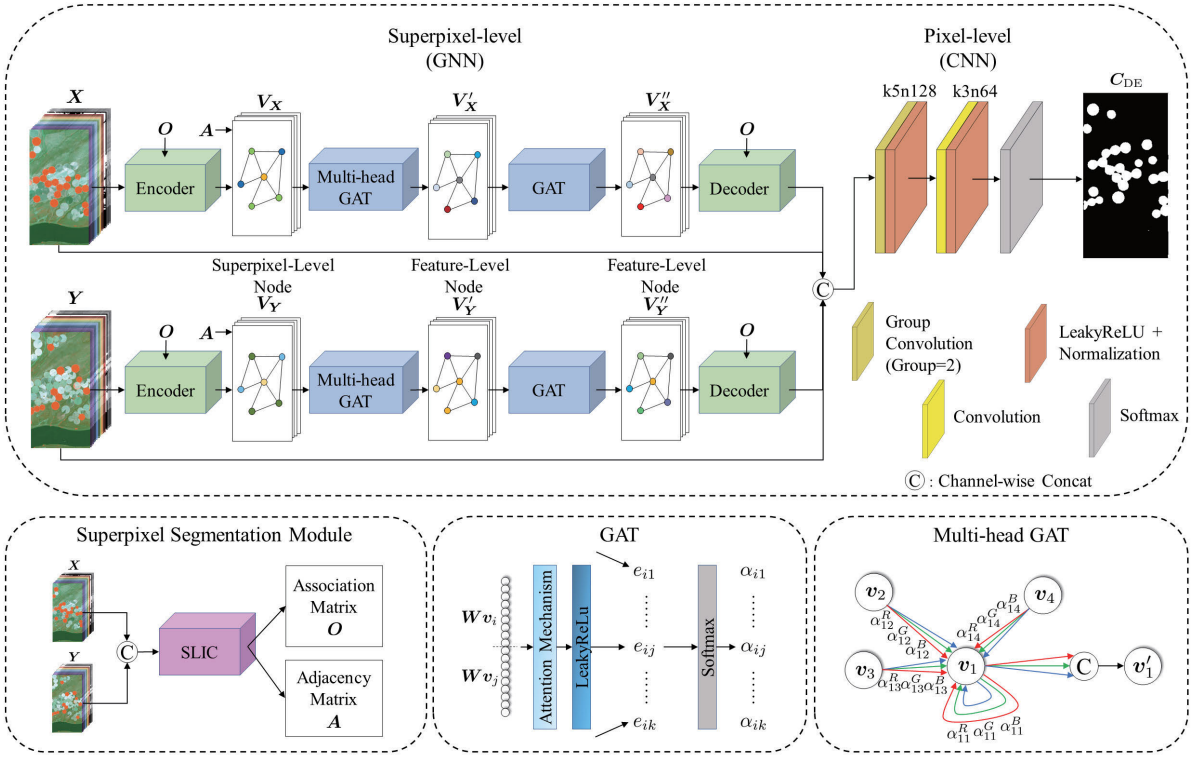


Fig. 3. Architecture of the deep feature distilling neural network.

neighboring superpixels. Finally, CNN layers fuse superpixel- and pixel-level features for generating the change map $\boldsymbol{C}_{\mathrm{DE}}$.

In the first stage, we need to convert input HSIs $\boldsymbol{X}$ and $\boldsymbol{Y}$ into graph structures to adopt GAT layers. Inspired by Dong et al. [37], it is achieved by partitioning $\boldsymbol{X}$ and $\boldsymbol{Y}$ into superpixels using the simple linear iterative clustering (SLIC) method [61]. Since SLIC is designed for the superpixel segmentation of RGB images, we first apply linear discriminant analysis (LDA) [62] to achieve dimension reduction on the bitemporal HSIs before conducting the segmentation; this is a commonly used trick to apply the seminal SLIC algorithm to analyze the HSIs [40]. A graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is then constructed from the spatially connected superpixels with $\mathcal{V}$ and $\mathcal{E}$ denoting the nodes and edges, respectively. To illustrate this process, we will use $\boldsymbol{X}$ to demonstrate the conversion of HSI into a graph structure and the propagation of information among the nodes; those for $\boldsymbol{Y}$ are similarly done. Let the set of the superpixels be $\{\mathcal{S}_i\}_{i=1}^{Z}$, where $\mathcal{S}_i$ denotes the ith superpixel and Z is the total number of the superpixels. The matrix form of the nodes $\mathcal{V}$ can be written as $\boldsymbol{V} = [\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_Z]^T \in \mathbb{R}^{Z\times M}$ with the ith node $\boldsymbol{v}_i$ being the mean of the pixels in the ith superpixel $\mathcal{S}_i$. The association matrix $\boldsymbol{O} \in \mathbb{R}^{L\times Z}$ is defined as

$$\mathbf{O}_{i,j} = \begin{cases} 1, & \text{if } \boldsymbol{x}_i \in \mathcal{S}_j, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

The graph encoder and decoder can then be represented as

$$\boldsymbol{V} = \text{Encoder}\ (\boldsymbol{X}, \boldsymbol{O}) = \tilde{\boldsymbol{O}}^T \boldsymbol{X}^T \tag{11}$$

$$\widehat{\boldsymbol{X}} = \text{Decoder}\ (\boldsymbol{V}, \boldsymbol{O}) = \boldsymbol{O}\boldsymbol{V} \tag{12}$$

where $\tilde{\boldsymbol{O}}_{i,j} = \boldsymbol{O}_{i,j} / \sum_{l=1}^{L} \boldsymbol{O}_{l,j}$. With the graph encoder and decoder, we can convert the bitemporal HSIs into graphs.

The process of constructing the hidden representations of each node from neighboring superpixels begins with the formation of the graph structure of the HSI. To accomplish this, the GAT module [41] is applied. A weight matrix $\boldsymbol{W}_1 \in \mathbb{R}^{N_1\times M}$ is utilized to provide higher level features of the HSIs via linear transformation followed by a self-attention element Att: $\mathbb{R}^{N_1} \times \mathbb{R}^{N_1} \mapsto \mathbb{R}$ computing attention coefficient e_{ij} as

$$e_{ij} = \text{Att}\big(\boldsymbol{W}_1\boldsymbol{v}_i, \boldsymbol{W}_1\boldsymbol{v}_j\big) \tag{13}$$

which reflects the importance of each element in the feature vector pair. The self-attention element highlights the relevant elements of the bitemporal features and can be implemented by a single-layer feedforward neural network with a trainable vector $\boldsymbol{a} \in \mathbb{R}^{2N_1}$ as

$$e_{ij} = \text{LeakyReLu}\big(\boldsymbol{a}^T\big(\boldsymbol{W}_1\boldsymbol{v}_i \| \boldsymbol{W}_1\boldsymbol{v}_j\big)\big) \tag{14}$$

where $\|$ is the concatenation operators and LeakyReLu denotes the leaky rectified linear unit activation function with a commonly used setting negative slope $\alpha_{\mathrm{GAT}} = 0.2$ for GAT [37], as illustrated in Fig. 3. To focus on the change between adjacent superpixels, the computation of e_{ij} is only performed for $j \in \mathcal{N}_i$, where $\mathcal{N}_i$ is the first-order neighborhood of i. The attention coefficients are then normalized through the application of the softmax function as

$$\alpha_{ij} = \text{softmax}_j\big(e_{ij}\big) = \frac{\exp\big(e_{ij}\big)}{\sum_{k\in\mathcal{N}_i}\exp(e_{ik})}. \tag{15}$$

Furthermore, to more effectively capture the information from the neighboring superpixels, the multi-head mechanism is

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employed to highlight the first-order neighborhood, leading to the calculation of the aggregated node, denoted as

$$\boldsymbol{v}_i' = \|_{h=1}^{H} \text{elu}\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij}^h \boldsymbol{W}_1^h \boldsymbol{v}_j\right) \tag{16}$$

where the subscript h denotes the hth attention mechanism, H is the amount of the attention mechanisms, and "elu" denotes the exponential linear unit (ELU) activation function. In order to enhance stability, deeper features at the superpixel level are extracted using a single-head GAT layer. Subsequently, the output node can be computed through the application of another GAT layer, which can be represented as

$$\boldsymbol{v}_i'' = \text{elu}\left(\sum_{j\in\mathcal{N}_i} \beta_{ij} \boldsymbol{W}_2 \boldsymbol{v}_j'\right) \tag{17}$$

where $\boldsymbol{W}_2 \in \mathbb{R}^{N_2\times N_1 H}$ and $\beta : \mathbb{R}^{N_2}\times\mathbb{R}^{N_2} \mapsto \mathbb{R}$ are the weight matrix and the attention mechanism, respectively. By implementing (17), the higher level features containing information of the second-order neighborhood are utilized after node information propagation. However, since this information is not as critical as that of the single-order neighborhood, the single-head GAT layer is sufficient, as illustrated in Fig. 3. Once the deep nodes have been learned at the superpixel level, the above decoder function is applied to convert them back to the original spatial size, resulting in the superpixel-level feature map, i.e.,

$$\boldsymbol{X}' = \text{ Decoder } \left(\boldsymbol{V}'', \boldsymbol{O}\right). \tag{18}$$

Although we have extracted the embedded feature at the superpixel level, it should be noted that the pixel-level information is still unavailable. To address it, we concatenate the superpixel-level feature maps with input HSIs, followed by a group convolution fusing the information embedded in the superpixel-level maps and the pixel-level input images (i.e., $\boldsymbol{X}$ and $\boldsymbol{Y}$). In order to prevent the occurrence of dead neurons, we then adopt the LeakyReLu activation function with a smaller value of $\alpha_{\text{CNN}} = 0.01$ for CNN to prevent the overfitting issue [63]. Finally, a convolution layer with the LeakyReLu activation function followed by a softmax function is introduced to generate the change map $\boldsymbol{C}_{\text{DE}}$. Let $\boldsymbol{C}_{\text{GT}}$ represent the ground-truth change map. Cross-entropy is adopted as the loss function, which is defined as

$$L(\boldsymbol{C}_{\text{GT}}, \boldsymbol{C}_{\text{DE}}) := -\frac{1}{T}\sum_{i=1}^{T}([\boldsymbol{c}_{\text{GT}}]_i \log([\boldsymbol{c}_{\text{DE}}]_i)) \tag{19}$$

where T is the total number of training samples and $[\boldsymbol{c}_{\text{GT}}]_i$ and $[\boldsymbol{c}_{\text{DE}}]_i$ are the ith training samples from $\boldsymbol{C}_{\text{GT}}$ and $\boldsymbol{C}_{\text{DE}}$, respectively.

As the change map $\boldsymbol{C}_{\text{DE}}$ learned from limited training samples is obtained, the DE regularization term extracts useful information from $\boldsymbol{C}_{\text{DE}}$ by the SFEFO $\|\cdot\|_{\boldsymbol{Q}}^2$. Owing to the fact that $\boldsymbol{C}_{\text{DE}}$ is a single band change map, the PSD matrix $\boldsymbol{Q}$ in $\|\cdot\|_{\boldsymbol{Q}}^2$ can be derived as 1-D PCA, which can be represented as the identity mapping. Consequently, the deep regularization can be reformulated as

Algorithm 1 CODE-HCD Algorithm for Solving (21)

1: **Given** $\lambda > 0$, $\mu > 0$. Initialize $\boldsymbol{c}^0 := \boldsymbol{0}$, $\boldsymbol{d}_i^0 := \boldsymbol{0}, \forall i \in \mathcal{I}_4$, and $q := 0$.
2: **repeat**
3: Update $\{\boldsymbol{c}_i^{q+1}\}_{i=1}^4 \in \arg\min_{\{\boldsymbol{c}_i\}} \mathcal{L}(\boldsymbol{c}^q, \{\boldsymbol{c}_i\}_{i=1}^4, \{\boldsymbol{d}^q\}_{i=1}^4)$;
4: Update $\boldsymbol{c}^{q+1} \in \arg\min_{\boldsymbol{c}} \mathcal{L}(\boldsymbol{c}, \{\boldsymbol{c}_i^{q+1}\}_{i=1}^4, \{\boldsymbol{d}^q\}_{i=1}^4)$;
5: Update $\boldsymbol{d}_i^{q+1} := \boldsymbol{d}_i^q + \boldsymbol{c}_i^{q+1} - \boldsymbol{c}^{q+1}, \ \forall i \in \mathcal{I}_4$;
6: $q := q + 1$;
7: **until** the predefined stopping criterion is met.
8: **Output** $\boldsymbol{c}^\star := \boldsymbol{c}^q$.

$\|\boldsymbol{c}-\boldsymbol{c}_{\text{DE}}\|_{\boldsymbol{Q}}^2 = (\boldsymbol{c}-\boldsymbol{c}_{\text{DE}})^T\boldsymbol{Q}(\boldsymbol{c}-\boldsymbol{c}_{\text{DE}}) = (\boldsymbol{c}-\boldsymbol{c}_{\text{DE}})^T\boldsymbol{I}_L(\boldsymbol{c}-\boldsymbol{c}_{\text{DE}}) = \|\boldsymbol{c}-\boldsymbol{c}_{\text{DE}}\|_2^2$. After completing the design of REG($\boldsymbol{c}$), the final convex HCD criterion can be reformulated as

$$\min_{\boldsymbol{c}\in[0,1]^L} \|\text{Diag}(\boldsymbol{1}_L - \boldsymbol{c})\psi\|_2^2 + \lambda\|\boldsymbol{c}-\boldsymbol{c}_{\text{DE}}\|_2^2. \tag{20}$$

Thus, the optimal solution $\boldsymbol{c}^\star$ can be obtained by solving (20). It is noteworthy that the convex criterion (20) can be efficiently solved by a convex optimizer due to its concise mathematical structure (see Section III-C).

C. HCD Algorithm Design and Implementation

Since the convex criterion (20) is large-scale, the ADMM optimizer [17], [18] is suitable to implement the HCD algorithm. Before this can be done, we first rewrite the data-fitting term $\|\text{Diag}(\boldsymbol{1}_L - \boldsymbol{c})\psi\|_2^2 = \|\psi\|_2^2 + \|\Phi\boldsymbol{c}\|_2^2 - 2\phi^T\boldsymbol{c}$ by defining $\Phi := \text{Diag}(\psi)$ and $\phi := \text{diag}(\psi\psi^T)$. This allows us to reformulate (20) into general ADMM form [17], [18] as

$$\min_{\boldsymbol{c}=\boldsymbol{c}_i, i=1,\ldots,4} \frac{1}{2}\|\Phi\boldsymbol{c}_1\|_2^2 - \phi^T\boldsymbol{c}_2 + \frac{\lambda}{2}\|\boldsymbol{c}_3 - \boldsymbol{c}_{\text{DE}}\|_2^2 + I_{\mathcal{C}}(\boldsymbol{c}_4) \tag{21}$$

where $\mathcal{C} := [0,1]^L$ on which the indicator function $I_{\mathcal{C}}(\boldsymbol{c}_4)$ has value 0 (otherwise, ∞). The augmented Lagrangian of (21) can be derived as

$$\begin{aligned}\mathcal{L}\left(\boldsymbol{c}, \{\boldsymbol{c}_i\}_{i=1}^4, \{\boldsymbol{d}\}_{i=1}^4\right) &= \frac{1}{2}\|\Phi\boldsymbol{c}_1\|_2^2 - \phi^T\boldsymbol{c}_2 + \frac{\lambda}{2}\|\boldsymbol{c}_3 - \boldsymbol{c}_{\text{DE}}\|_2^2 + I_{\mathcal{C}}(\boldsymbol{c}_4) \\ &\quad + \sum_{i=1}^{4}\frac{\mu}{2}\|\boldsymbol{c} - \boldsymbol{c}_i - \boldsymbol{d}_i\|_2^2\end{aligned}$$

where $\mu > 0$ denotes the penalty parameter and $\{\boldsymbol{d}_i\} \in \mathbb{R}^L$ are the scaled dual variables. The algorithm for solving (21) is detailed in Algorithm 1, where q represents the iteration number. The convergence guarantee of CODE-HCD is stated in Theorem 1, whose proof can be found in the Appendix.

Theorem 1: As long as $\mu > 0$, every limit point of the sequence $\{\boldsymbol{c}^q\}$ generated by Algorithm 1 must be an optimal solution of the HCD problem (21). □

To implement Algorithm 1, we need to derive $\{\boldsymbol{c}_i^{q+1}\}_{i=1}^4$ and $\boldsymbol{c}^{q+1}$. Thanks to the concise mathematical form of (21), the closed-form solutions can all be derived with high computational efficiency. First, we handle $\boldsymbol{c}_1^{q+1}$, which can be

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reformulated as

$$\boldsymbol{c}_1^{q+1} = \arg\min_{\boldsymbol{c}_1} \frac{1}{2}\|\Phi \boldsymbol{c}_1\|_2^2 + \frac{\mu}{2}\left\|\boldsymbol{c}^q - \boldsymbol{c}_1 - \boldsymbol{d}_1^q\right\|_2^2$$
$$= \mu\left(\Phi^T\Phi + \mu \boldsymbol{I}_L\right)^{-1}\left(\boldsymbol{c}^q - \boldsymbol{d}_1^q\right) \quad (22)$$
$$= \mu\left[\left(\psi_1^2 + \mu\right)^{-1}, \ldots, \left(\psi_L^2 + \mu\right)^{-1}\right]^T \odot \left(\boldsymbol{c}^q - \boldsymbol{d}_1^q\right). \quad (23)$$

Unlike (22), solution (23) does not involve the inverse of the large $L \times L$ matrix, making the computation of $\boldsymbol{c}_1$ efficient. Second, $\boldsymbol{c}_2^{q+1}$ can be simplified as

$$\boldsymbol{c}_2^{q+1} = \arg\min_{\boldsymbol{c}_2} -\phi^T \boldsymbol{c}_2 + \frac{\mu}{2}\left\|\boldsymbol{c}^q - \boldsymbol{c}_2 - \boldsymbol{d}_2^q\right\|_2^2$$
$$= \frac{\phi}{\mu} + \left(\boldsymbol{c}^q - \boldsymbol{d}_2^q\right)$$

which only involves vector additions resulting in a computationally efficient solution. Third, we have $\boldsymbol{c}_3^{q+1}$, which can be written as

$$\boldsymbol{c}_3^{q+1} = \arg\min_{\boldsymbol{c}_3} \frac{\lambda}{2}\|\boldsymbol{c}_3 - \boldsymbol{c}_{\mathrm{DE}}\|_2^2 + \frac{\mu}{2}\left\|\boldsymbol{c}^q - \boldsymbol{c}_3 - \boldsymbol{d}_2^q\right\|_2^2$$
$$= \frac{\lambda \boldsymbol{c}_{\mathrm{DE}} + \mu\left(\boldsymbol{c}^q - \boldsymbol{d}_3^q\right)}{\lambda + \mu}.$$

The solution of $\boldsymbol{c}_3^{q+1}$ consists of a simple mathematical form yielding a fast computational speed. Then, the subproblem of $\boldsymbol{c}_4^{q+1}$ can be represented as

$$\boldsymbol{c}_4^{q+1} = \arg\min_{\boldsymbol{c}_4} I_{\mathcal{C}}(\boldsymbol{c}_4) + \frac{\mu}{2}\left\|\boldsymbol{c}^q - \boldsymbol{c}_4 - \boldsymbol{d}_2^q\right\|_2^2$$
$$= \arg\min_{\boldsymbol{c}_4 \in \mathcal{C}}\left\|\boldsymbol{c}^q - \boldsymbol{c}_4 - \boldsymbol{d}_2^q\right\|_2^2$$

whose closed-form solution is nothing but the projection of $\boldsymbol{c}^q - \boldsymbol{d}_4^q$ onto $\mathcal{C}$, which can be represented as

$$\left[\boldsymbol{c}_4^{q+1}\right]_\ell = \begin{cases} \left[\boldsymbol{c}^q - \boldsymbol{d}_4^q\right]_\ell, & \text{if } \left[\boldsymbol{c}^q - \boldsymbol{d}_4^q\right]_\ell \in [0,1], \\ 0, & \text{if } \left[\boldsymbol{c}^q - \boldsymbol{d}_4^q\right]_\ell < 0, \\ 1, & \text{otherwise}, \end{cases} \quad \forall \ell \in \mathcal{I}_L.$$

This closed-form solution only involves simple operations and is, hence, fast. Finally, the problem solving $\boldsymbol{c}^{q+1}$ can be redefined as

$$\boldsymbol{c}^{q+1} = \arg\min_{\boldsymbol{c}} \mathcal{L}\left(\boldsymbol{c}, \left\{\boldsymbol{c}_i^{q+1}\right\}_{i=1}^4, \left\{\boldsymbol{d}^q\right\}_{i=1}^4\right)$$
$$= \frac{1}{4}\sum_{i=1}^4\left(\boldsymbol{c}_i^{q+1} + \boldsymbol{d}_i^q\right).$$

This completes the algorithm design of CODE-HCD, whose implementation is computationally efficient. The superiority of the proposed CODE-HCD will be experimentally demonstrated in Section IV.

IV. Experimental Results

The lack of labeled datasets for HCD problems makes it crucial and practical to develop methods for solving HCD problems with limited training labels [24]. In this section, we aim to demonstrate the superiority of the proposed CODE-HCD under low sampling rate conditions by comparing it with state-of-the-art CD algorithms. Specifically, in Section IV-A, we provide a brief overview of three real hyperspectral datasets and two quantitative assessments. In Section IV-B, we investigate the performance of the CD algorithms in a practical scenario where a fixed limited sampling rate is used. The effect of different sampling rates on DE-based CD algorithms is discussed in Section IV-C. Moreover, the sensitivity of the parameters of the CO algorithm and the DE model is discussed in Section IV-D. The computational efficiency of the CD algorithms is analyzed in Section IV-E. In Section IV-F, an ablation study is conducted. Finally, we investigate the statistical behavior analysis for the proposed CODE-HCD using one additional dataset of similar statistical property in Section IV-G.

A. Experiment Setting

In this section, we compare the proposed CODE-HCD with benchmark CD methods, including CVA [5], PCA [12], SVMs [13], TDRD [15], CD-ADMM [16], GETNET [24], and CSDBF [40]. These methods comprise a mix of unsupervised and supervised algorithms, with CVA, PCA, TDRD, and CD-ADMM being unsupervised, and the others being supervised. It is noteworthy that the methods of CVA, TDRD, CD-ADMM, and the proposed CODE-HCD necessitate the selection of an appropriate threshold value, and the k-means is adopted to classify the changed pixels and unchanged pixels for these methods in our experiment.

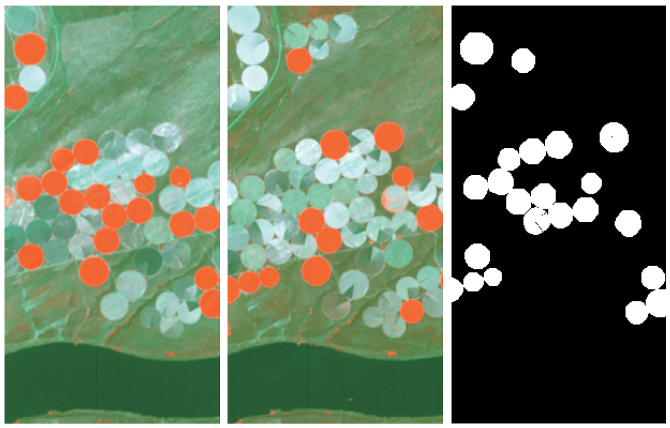
In order to assess the performance of the proposed CODE-HCD algorithm, we use three widely studied real HSI datasets. Each dataset includes two bitemporal HSIs and a ground-truth image, as will be discussed in the following. The two bitemporal HSIs cover the same spatial region and were captured at different dates, while the ground-truth images record the information on the distribution of the changed and unchanged pixels. The three benchmark datasets are introduced as follows.

1) Hermiston Dataset: The dataset was acquired by the Hyperion sensor over Hermiston city in Umatilla County, United States, on May 1, 2004, and May 8, 2007, with 242 spectral bands ranging from 350 to 2580 nm [64]. Bitemporal HSIs containing $L = 390 \times 200$ pixels with $M = 154$ after noise elimination are used in the experiment. The changed areas of this dataset are mainly agricultural change, including the transitions among crops, soil, water, and others. The false color images of the bitemporal Hermiston dataset and the ground-truth change map are illustrated in Fig. 4.

2) Jiangsu Dataset: The dataset was acquired by the Hyperion sensor over Jiangsu province, China, on May 3, 2013, and December 31, 2013, with 242 spectral bands ranging from 350 to 2580 nm [24]. Bitemporal HSIs containing $L = 463 \times 241$ pixels with $M = 128$ after noise elimination are used in the experiment. The changed areas of this dataset are mainly the disappearance of the substance in the river. The false color images of the bitemporal Jiangsu dataset and the ground-truth change map are illustrated in Fig. 5.

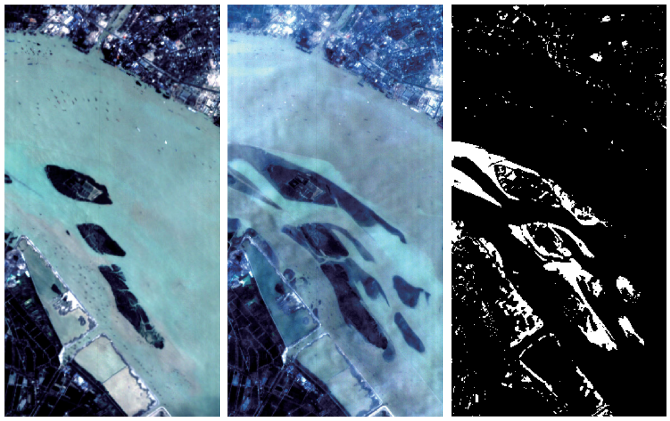
3) Yancheng Dataset: The dataset was acquired by the Hyperion sensor over a wetland agricultural land in Yancheng, Jiangsu province, China, on May 3, 2006, and April 23, 2007, with 242 spectral bands ranging from 350 to 2580 nm [65]. Bitemporal HSIs containing $L = 450 \times 140$ pixels with

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(a) (b) (c)

Fig. 4. False color images of the Hermiston dataset. (a) HSI acquired on May 1, 2004. (b) HSI acquired on May 8, 2007. (c) Ground-truth change map.



(a) (b) (c)

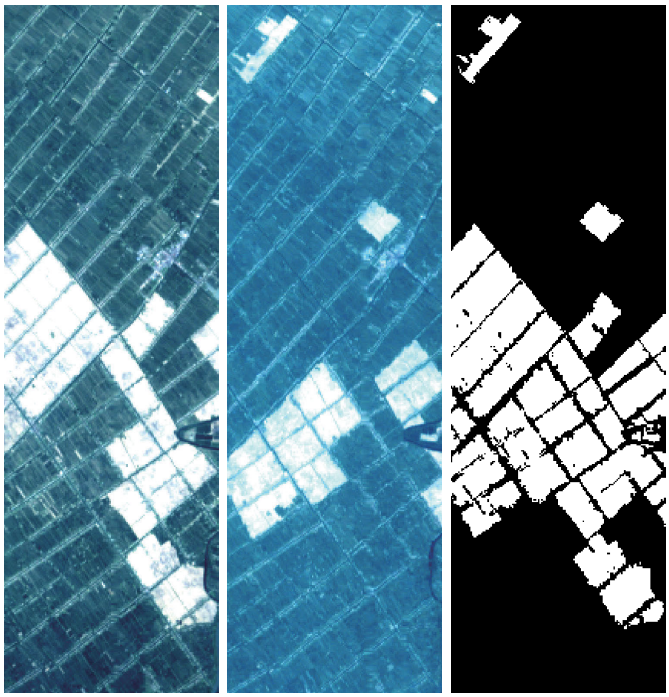
Fig. 5. False color images of the Jiangsu dataset. (a) HSI acquired on May 3, 2013. (b) HSI acquired on December 31, 2013. (c) Ground-truth change map.

$M = 155$ after noise elimination are used in the experiment. The changed areas of this dataset are mainly agricultural changes in vegetation, bare land, water, and soil [66]. The false color images of the bitemporal Yancheng dataset and the ground-truth change map are illustrated in Fig. 6.

For a fair comparison, four widely used assessments, including overall accuracy (OA), Cohen's kappa coefficient κ, precision indicator, and recall indicator, are adopted in this article to quantitatively evaluate the performance of the CD algorithms. OA is a measure of the ratio of the number of correctly classified pixels to the total pixel number, defined as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$

where TP, TN, FP, and FN denote true positive, true negative, false positive, and false negative, respectively. The larger the value of OA, the more accurate the final change map is. Cohen's kappa coefficient κ is a more robust measure compared to OA because it takes into account the agreement between the output change map and the ground-truth image.



(a) (b) (c)

Fig. 6. False color images of the Yancheng dataset. (a) HSI acquired on May 3, 2006. (b) HSI acquired on April 23, 2007. (c) Ground-truth change map.

The definition of Cohen's kappa coefficient can be written as

$$\kappa = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}}$$

where PRE is the hypothetical probability of chance agreement, which can be computed as

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN}) + (\mathrm{FN} + \mathrm{TN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}.$$

A higher value of κ indicates better accuracy in the CD result. Moreover, the precision indicator is introduced to investigate the rate of TP of the positive results, which can be derived as

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}.$$

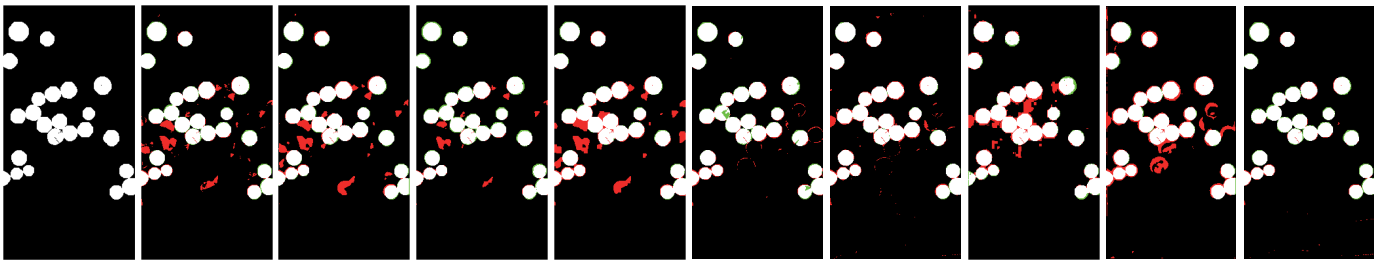
A higher value of precision indicator corresponds to superior accuracy of the positive results. On the other hand, the recall indicator tests the ability of the CD algorithms to detect the changed pixels with its mathematical expression being

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}.$$

A higher value of the recall indicator implies that the algorithm has a greater capacity for correctly identifying those changed pixels in the dataset.

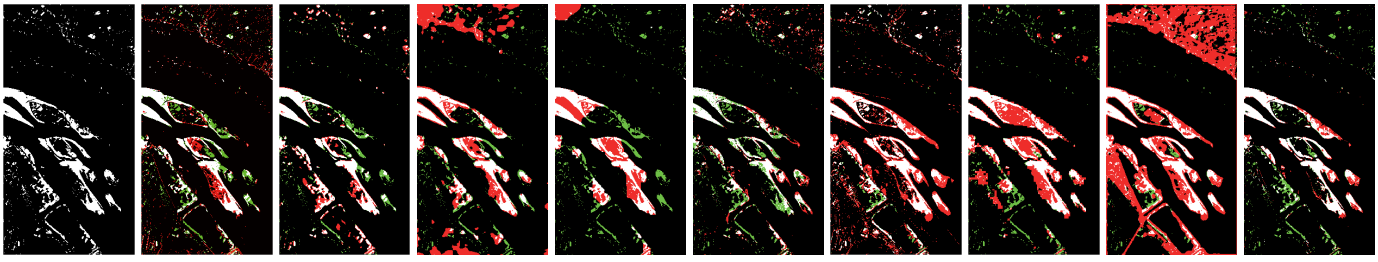
In accordance with our research philosophy, which posits that software engineers should strive for a unified parameter setting in order to achieve a practical and user-friendly algorithm as previously demonstrated in our works, such as [49], we adopt the parameter setting $(\lambda, \mu) = (10, 0.001)$ for the proposed CODE-HCD in this article. In addition,

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 7. CD results of the CD algorithms on the Hermiston dataset with FN and FP marked in green and red, respectively. (a) Ground truth. (b) CVA. (c) PCA. (d) TDRD. (e) CD-ADMM. (f) SVMs. (g) GETNET. (h) CSDBF. (i) GACDN. (j) CODE-HCD.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. CD results of the CD algorithms on the Jiangsu dataset with FN and FP marked in green and red, respectively. (a) Ground truth. (b) CVA. (c) PCA. (d) TDRD. (e) CD-ADMM. (f) SVMs. (g) GETNET. (h) CSDBF. (i) GACDN. (j) CODE-HCD.

we have established a parameter setting of $(N_1, N_2, H) = (64, 64, 2)$ for the GAT modules in the GACDN. The learning rate for training the GACDN model is set at 0.0001, utilizing the Adam optimizer [67] with a decay rate of 0.9 every 10 epochs, and the maximum number of epochs is set at 250. The superpixel segmentation scale s, which determines the number of superpixels $Z = (L/s)$, is set as 60 during the superpixel-generating procedure. Based on the natural that the changed pixels are typically fewer than the unchanged pixels, all of the training data in this article were sampled with a ratio of positive samples to negative samples of 1:2, which is also adopted in [24], and then divided into a training set and a validation set with a ratio of 7:3.

B. Performance Under a Low Training Sampling Rate

As mentioned in Section I, the acquisition and labeling of training samples of HSIs from real-world scenarios are both time-consuming and expensive, motivating us to investigate the performance of benchmark CD algorithms under a limited training sampling rate. In this experiment, a sampling ratio of 0.1% is applied to all three datasets. For a fair comparison, qualitative and quantitative results are discussed to evaluate the performance of the CD methods. The qualitative results are shown in Figs. 7–9, with FN and FP marked in green and red, respectively. The quantitative results are shown in Table I, with the best performance marked in an *underlined boldfaced number*, while the second marked just using a *boldfaced number*.

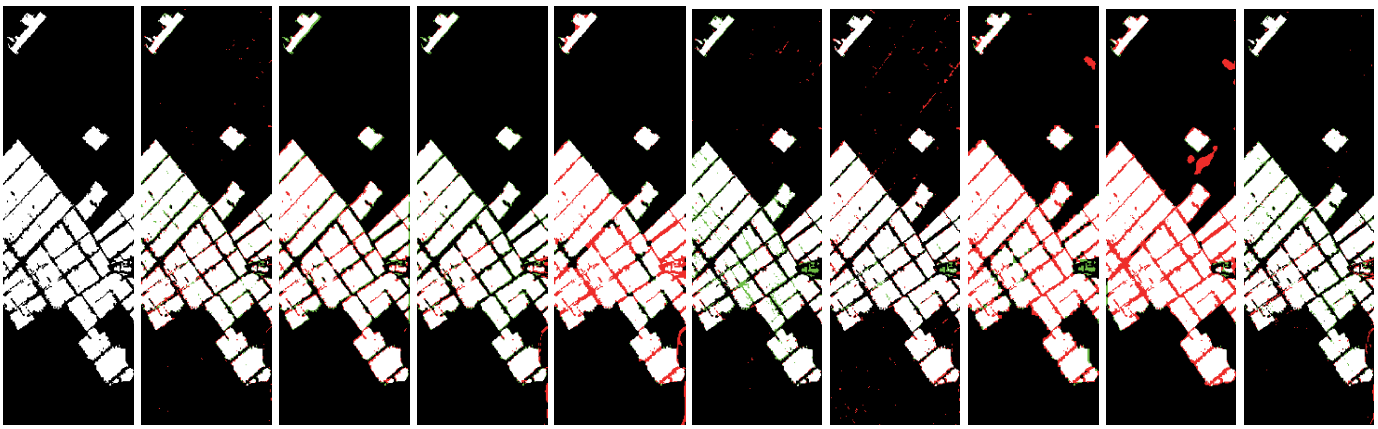
1) Experiment on the Hermiston Dataset: The qualitative results of the CD algorithms on the Hermiston Dataset are shown in Fig. 7. As can be observed in Fig. 7(b)–(e), there are artifacts in the output change maps. This is due to the fact that these methods are unable to learn deep features for the specific scenario. For CD-ADMM, as illustrated in Fig. 7(e), determining the similarity by merely computing the Euclidean distance has neglected the effect of spectral variation, resulting in unstable performance, as discussed in Section III-A. On the other hand, as shown in Fig. 7(f)–(i), the output change maps of supervised methods deviate significantly from the ground-truth map, particularly in terms of false positives, likely as a result of the limited number of training samples. However, despite the presence of false positive pixels in $\boldsymbol{C}_{\mathrm{DE}}$ in Fig. 7(i), the convex HCD criterion (9) successfully extracts useful features from $\boldsymbol{C}_{\mathrm{DE}}$ through the proposed deep regularizer and detects the changed area with a high degree of similarity to the ground-truth map presented in Fig. 7(a). As can be seen in Table I, the κ coefficients of the unsupervised HCDs are generally lower than those of supervised ones, which is consistent with the observations made in the qualitative analysis. The proposed CODE-HCD exhibits exceptional results in terms of OA and κ and stands at the forefront when judged through both qualitative and quantitative comparisons.

2) Experiment on the Jiangsu Dataset: The qualitative results of the CD algorithms on the Jiangsu dataset are shown in Fig. 8. CVA can roughly detect the changed pixels but still produce false negatives, while TDRD tends to generate false positives, as depicted in Fig. 8(b) and (d). In addition, PCA produces a good result, as seen in Fig. 8(c); however, it fails to capture the details of the changed areas. As can be seen in Fig. 8(f), SVMs are able to detect some true positives and true negatives but still suffer from the lack of training samples. In Fig. 8(f), CD-ADMM tends to detect continuously changed areas causing the false positives due to its TV regularization scheme that promotes the piecewise smoothness of the change map, which does not align with the scattered distribution nature of the changed pixels in this dataset. Comparing the

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TABLE I
QUANTITATIVE COMPARISON ON THE FOUR REAL HYPERSPECTRAL DATASETS

		Unsupervised method				Semi-supervised method				
Dataset	Index	CVA	PCA	TDRD	CD-ADMM	SVMs	GETNET	CSDBF	GACDN	CODE-HCD
Hermiston	OA(↑)	0.970	0.965	0.963	0.960	0.974	**0.982**	0.942	0.864	**0.986**
	κ(↑)	0.871	0.855	0.849	0.839	0.879	**0.920**	0.765	0.686	**0.938**
	Precision(↑)	0.839	0.810	0.791	0.776	0.936	**0.957**	0.898	0.622	**0.947**
	Recall(↑)	0.944	0.951	0.966	**0.968**	0.857	0.908	0.720	**0.984**	0.944
Jiangsu	OA(↑)	0.934	**0.948**	0.881	0.916	0.934	0.923	0.911	0.797	**0.954**
	κ(↑)	0.602	**0.676**	0.448	0.511	0.532	0.639	0.551	0.299	**0.707**
	Precision(↑)	0.614	0.701	0.397	0.512	0.636	**0.916**	0.733	0.260	**0.747**
	Recall(↑)	0.664	0.708	0.708	0.613	0.547	0.545	0.516	**0.785**	**0.719**
Yancheng	OA(↑)	**0.969**	0.955	0.966	0.940	0.950	0.964	0.931	0.923	**0.970**
	κ(↑)	**0.926**	0.892	0.917	0.861	0.879	0.916	0.837	0.827	**0.927**
	Precision(↑)	0.942	0.910	0.931	0.836	0.917	**0.979**	0.917	0.806	**0.956**
	Recall(↑)	0.954	0.938	0.952	**0.984**	0.911	0.910	0.859	**0.978**	0.941
Bay Area	OA(↑)	0.890	0.894	0.904	**0.920**	0.796	0.916	0.913	0.869	**0.934**
	κ(↑)	0.761	0.770	0.800	**0.832**	0.514	0.829	0.813	0.726	**0.856**
	Precision(↑)	0.896	0.888	0.827	0.857	**0.962**	0.931	0.863	0.796	**0.976**
	Recall(↑)	0.800	0.819	**0.940**	**0.942**	0.474	0.883	0.910	0.885	0.845



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. CD results of the CD algorithms on the Yancheng dataset with FN and FP marked in green and red, respectively. (a) Ground truth. (b) CVA. (c) PCA. (d) TDRD. (e) CD-ADMM. (f) SVMs. (g) GETNET. (h) CSDBF. (i) GACDN. (j) CODE-HCD.

DE methods illustrated in Fig. 8(g)–(i), the outputs of CSDBF and GACDN lose detailed information mainly as a result of the superpixel-level learning with small training samples. On the other hand, GETNET detects the changed areas more precisely but still generates false positives. The proposed CODE-HCD is able to accurately find the changed pixels, as depicted in Fig. 8(j), even though the change map of GACDN only roughly identifies changed areas. By applying the deep regularizer instead of the traditional ones, such as TV, the proposed CODE-HCD has better adaptability on scattered changed pixels than CD-ADMM. The quantitative assessments presented in Table I support the observations made in the qualitative analysis, indicating that CODE-HCD has the best performance in terms of both OA and κ, while SVMs have weak scores on the two metrics.

3) Experiment on the Yancheng Dataset: The qualitative results of the CD algorithms on the Yancheng dataset are shown in Fig. 9. As can be observed in Fig. 9(b), (c), and (e), the unsupervised methods demonstrate good performances, with the only drawback being the lack of detail in the change maps produced by PCA and TDRD. Meanwhile, CD-ADMM struggles with detecting the boundaries of the changed regions, as shown in Fig. 9(f). As illustrated in Fig. 9(d), there are false negatives in the change map produced by the SVMs, which can be observed by the recall indicator, likely due to the limited number of training samples. While the DE algorithms are better able to handle the limited training data situation, the outputs of CSDBF and GACDN contain false positives at the gaps between changed regions, as depicted in Fig. 9(h) and (i), respectively. In addition, GETNET fails to detect about half of the changed part, as illustrated in Fig. 9(g). However, the proposed CODE-HCD algorithm is able to learn useful information from GACDN and to nearly perfectly generate change maps, which are almost identical to the ground-truth map, as demonstrated in Fig. 9(j). Compared to traditional DE methods, CODE-HCD can detect

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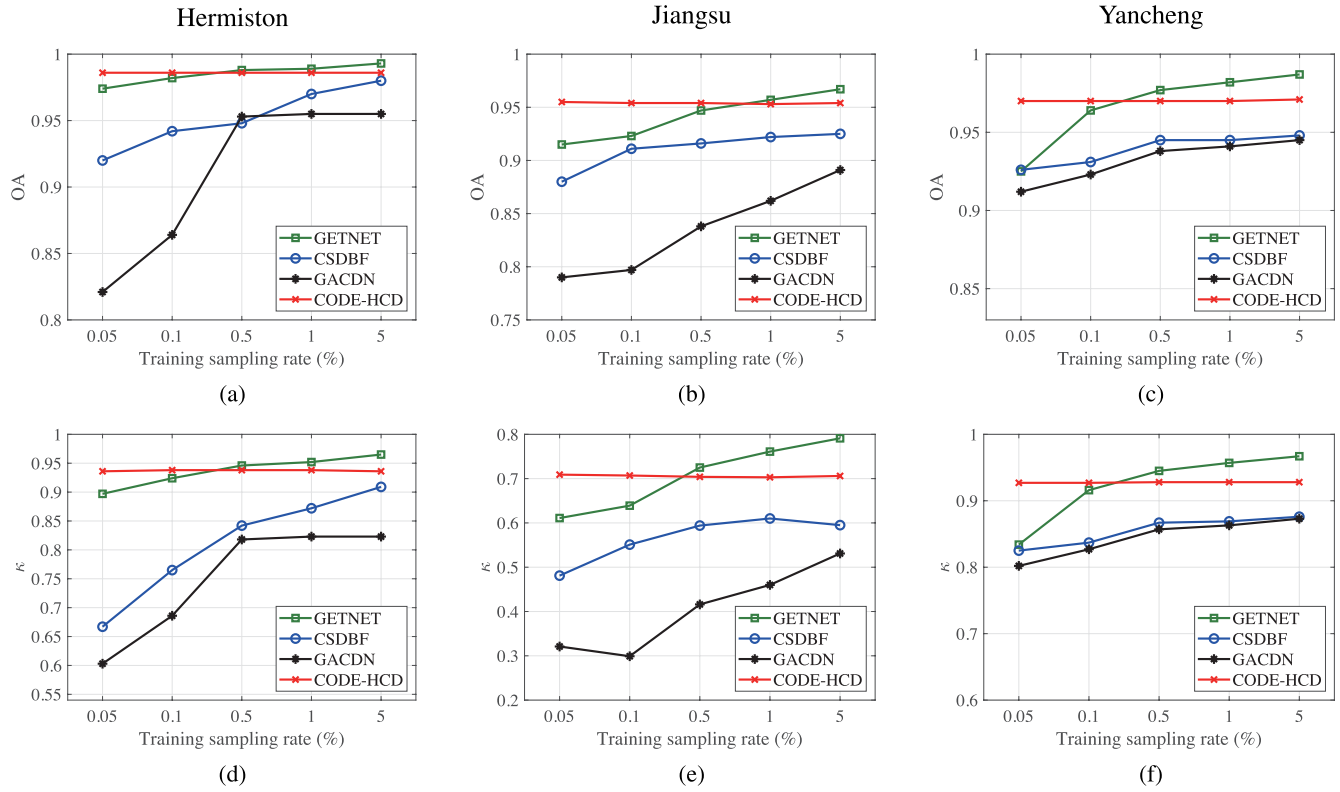


Fig. 10. Performance comparison in terms of OA and Cohen's kappa coefficient κ for different sampling rates (i.e., labeling rates) using the Hermiston dataset, the Jiangsu dataset, and the Yancheng dataset. (a) OA for the Hermiston dataset. (b) OA for the Jiangsu dataset. (c) OA for the Yancheng dataset. (d) κ for the Hermiston dataset. (e) κ for the Jiangsu dataset. (f) κ for the Yancheng dataset.

the details of the changed regions by introducing information on spectral similarity (i.e., SAM), especially under the scenario of quite limited training samples. As can be seen in Table I, CODE-HCD leads the CD algorithms in terms of quantitative assessments, while the performance of the supervised methods lags behind those of the unsupervised methods.

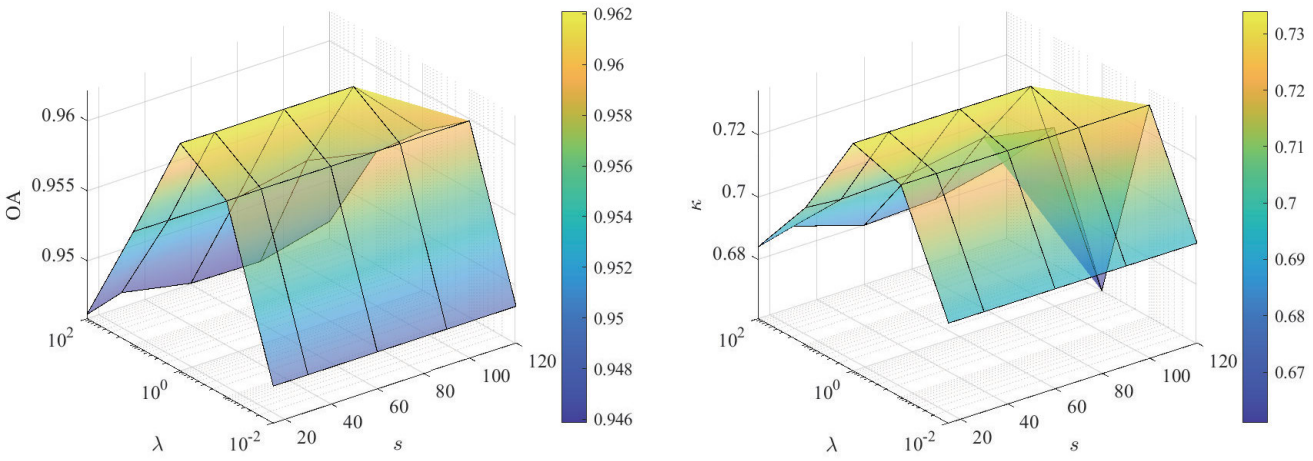
In the experiments conducted under conditions of a low training sampling rate, it can be seen that the performance of the supervised algorithms is highly affected. In contrast, traditional unsupervised methods demonstrate competitive performance on real-world hyperspectral datasets. The performance of CD-ADMM is not stable across different datasets, as its regularizer that promotes piecewise smooth change patterns does not align with the scattered distribution of changed pixels in some real-world scenarios, such as the Jiangsu dataset. This reveals the drawback of traditional CO-based criteria with customized regularizers that would not be directly applicable to new scenarios or datasets. The proposed CODE-HCD addresses this limitation by incorporating DE into the CO criterion, thereby leading to a scene-adaptive regularization scheme. GACDN can roughly detect the changed areas, while the proposed regularization term learns useful information from the rough solution to guide the data-fitting term in (9). Though CODE-HCD only slightly outperforms other CD methods on the Yancheng dataset, CODE-HCD has achieved state-of-the-art performance with significant improvements for other datasets, showing the stability of the proposed HCD algorithm. For example, in terms of κ, CODE-HCD significantly outperforms the strong peer method CVA with about 8% improvement for the Hermiston dataset and about 17% improvement for the Jiangsu dataset. The results of both quantitative and qualitative comparisons demonstrate the superiority of the proposed CODE-HCD in practical experimental conditions under limited training data.

C. Effect of Different Sampling Rate

As one can know that DE typically requires big data to achieve state-of-the-art performance, however, acquiring ample labeled data of HSIs can be time-consuming and costly. Therefore, algorithms that can effectively learn from small data are both essential and practical for remote sensing applications. In this section, we investigate the effect of the training sampling rate on DE methods and the proposed CODE-HCD. To this end, we conduct an experiment on the three hyperspectral datasets used in Section IV-B, with labeling rates ranging from 0.05% to 5%.

The results of the experiment involving the supervised DE methods are presented in Fig. 10. As can be observed, neural networks with more complex structures (i.e., GETNET and CSDBF) demonstrate better performance than the simple network structure (i.e., GACDN). The proposed CODE-HCD algorithm leverages the useful features embedded in the output of the GACDN, resulting in significant performance improvements. As the sampling rate increases, the performances of CSDBF and GETNET also improve, benefiting from the increased amount of training data. Conversely, though the simple network structure of GACDN limits its performance to

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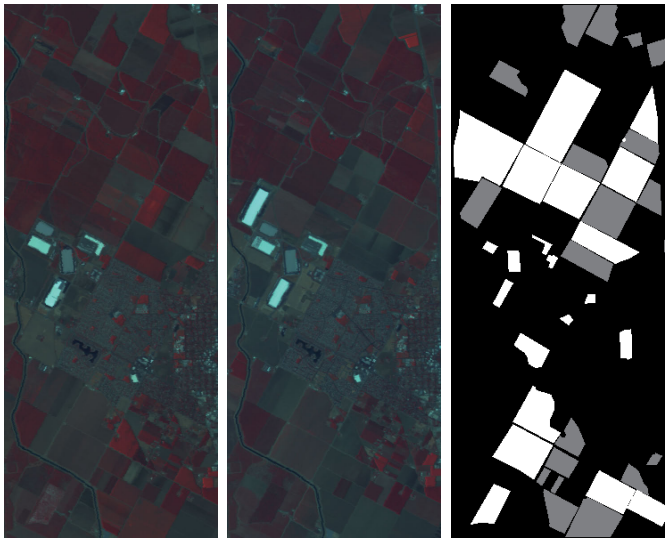
(a) (b)

Fig. 11. Performance comparison in terms of OA and Cohen's kappa coefficient κ on different parameter values of λ and s using the Jiangsu dataset. (a) OA. (b) κ.

some extent, CODE-HCD still produces competitive results. Therefore, to achieve excellent CD results in a challenging situation, the CODE-HCD criterion requires a stable output from a DE model, even if the model's structure is simple, as in the case of GACDN. This confirms the effectiveness of the CODE small-data learning theory [42] that is originally customized for hyperspectral satellite remote sensing.

D. Sensitivity Analysis

There are three key parameters in the proposed CODE-HCD, which are the super-pixel scale (i.e., the superpixel segmentation scale s), λ, and μ. Since μ simply controls the convergence speed without affecting the performance, we investigate the sensitivity analysis for λ and s; note that any setting of μ ensures the convergence as long as $\mu > 0$, as stated in Theorem 1. We perform an experiment on the Jiangsu dataset by varying λ over a range of values $\lambda \in \{10^n \mid n = -2, -1, 0, 1, 2\}$ and s over a range of values $s \in \{15, 30, 60, 90, 120\}$. The results, as depicted in Fig. 11, indicate that the superpixel segmentation scale has a negligible impact on the performance of the proposed HCD-CODE. Conversely, HCD-CODE is more sensitive to λ and yields the best performance around at $\lambda \in [1, 10]$ on the Jiangsu dataset, well echoing the default unifying parameter setting presented in Section IV-A. These findings reveal the potential that optimizing the parameters for each dataset could further upgrade the performance of CODE-HCD, but, in practice, this is less meaningful due to the lack of ground-truth change maps when considering new datasets. Though our results indicate that a more tailored parameter selection approach could improve the model's performance in specific scenarios, the unified setting still provides stable performance and is more practical when we have new datasets to test. In the future, a fully automatic regularization parameter setting of λ may also be investigated though it could be a highly challenging research line in general [68].



(a) (b) (c)

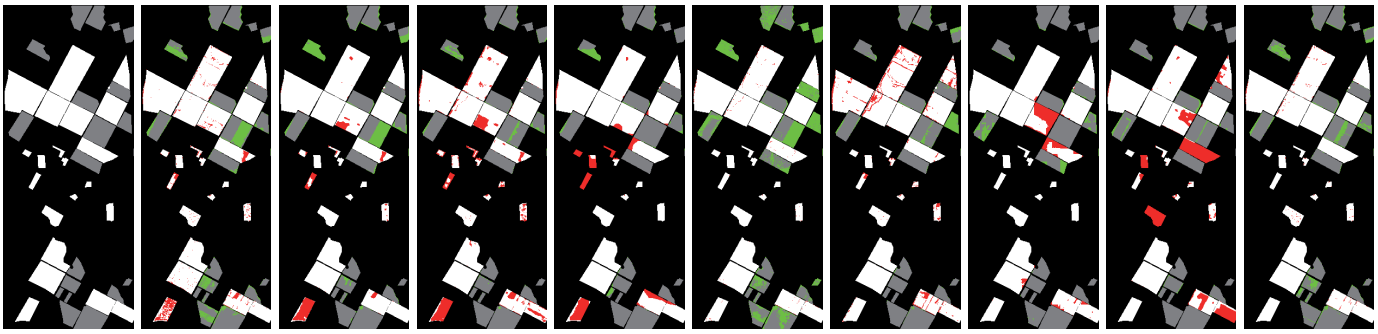
Fig. 12. False color images of the Bay Area dataset. (a) HSI acquired in 2013. (b) HSI acquired in 2015. (c) Ground-truth change map.

E. Computational Efficiency of HCD Criterion

In the proposed CODE-HCD, a convex criterion with a mathematically simplified regularization form has been employed to yield a computationally efficient algorithm. In this section, we compare the computational efficiency of the proposed CODE-HCD to the benchmark CD methods. All the experiments are conducted on the computer facility equipped with NVIDIA RTX-3090 GPU and AMD Ryzen 9 5950X CPU with 3.40-GHz speed and 128-GB random access memory. GACDN is carried out with Python-3.8.10 and Pytorch-1.7.0. CODE-HCD is implemented on Mathworks MATLAB R2021b.

The results of this comparison are summarized in Table II. The slow computational time of DE methods mainly comes from the semi-supervised scheme, which has to perform iterative testing and training in the meanwhile. CD-ADMM

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 13. CD results of the CD algorithms on the Bay Area dataset with FN and FP marked in green and red, respectively. (a) Ground truth. (b) CVA. (c) PCA. (d) TDRD. (e) CD-ADMM. (f) SVMs. (g) GETNET. (h) CSDBF. (i) GACDN. (j) CODE-HCD.

TABLE II

COMPARISON OF COMPUTATIONAL EFFICIENCY (i.e., RUNNING TIME) FOR BENCHMARK CD ALGORITHMS AND THE PROPOSED CODE-HCD (IN SECONDS)

Methods	Hermiston	Jiangsu	Yancheng
CVA	0.09	0.10	0.07
PCA	3.79	5.41	3.33
TDRD	4.41	5.33	3.76
CD-ADMM	146.21	635.06	120.19
SVMs	9.47	18.60	9.55
GETNET	53.01	94.16	45.51
CSDBF	95.46	184.54	68.44
GACDN	40.50	53.11	33.26
CODE-HCD	40.82	53.56	33.54

performs large-scale CO, so it also takes more time. Other methods involve relatively simple operators (e.g., averaging in k-means), so they are much faster. Also, the computational time is roughly proportional to the number of pixels. It is noteworthy that, despite the inclusion of iteratively updating procedures in the DE part, the ADMM algorithm (i.e., the CO part) of the proposed CODE-HCD can be executed within one second, attesting to its computational efficiency thanks to the simple mathematical form of the $\boldsymbol{Q}$-quadratic regularization.

F. Ablation Study

In this experiment, we conduct an ablation analysis to investigate the impact of different components of our proposed model. Specifically, we examined the effects of different similarity maps Ψ in the data-fitting term DF($\boldsymbol{c}$) and the deep information $\boldsymbol{C}_{\text{DE}}$ in the deep regularizer REG($\boldsymbol{c}$). The experiment results are shown in Table III, indicating that SAM (which measures spectral similarity) is crucial for achieving high accuracy, and notably, replacing SAM with the typical Euclidean data-fitting does result in significant performance degradation, especially in terms of κ. In addition, learning the deep information $\boldsymbol{C}_{\text{DE}}$ from another alternative network structure (i.e., GACDN) still leads to similar performance as the relatively sophisticated network (i.e., GETNET), echoing the aforementioned advantage that a simple neural network structure would still be sufficient to yield good performance under the CODE theory (cf. Section II-A). Our ablation study

TABLE III

ABLATION STUDY ON THE HERMISTON AND JIANGSU DATASETS IN TERMS OF OA AND κ INDICES

Models		Hermiston		Yancheng	
Ψ (data-fitting term)	$\boldsymbol{C}_{\text{DE}}$ (regularizer term)	OA	κ	OA	κ
Euclidean	GACDN	0.982	0.917	0.954	0.887
SAM	GACDN	**0.986**	**0.938**	**0.970**	**0.927**
SAM	GETNET	**0.986**	0.936	0.968	0.921

provides valuable insights into the design of the proposed CODE-HCD and shows the effectiveness of the proposed components in the algorithm.

G. Statistical Behavior Analysis

In this section, we conduct an experiment to investigate the statistical behavior of the proposed CODE-HCD for using another dataset with similar statistical characteristics. We select a subscene of the well-known Bay Area dataset, whose geometry characteristics (i.e., regular/periodic geometry structure) are similar to the Yancheng dataset. We consider the geometry shape as the key statistical characteristic because geometry shape is highly critical to the adopted superpixel mechanism. Both datasets are with changed regions both exhibiting shapes that are nearly regular. The dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Patterson in California, United States, in 2013 and 2015, with 242 spectral bands [40]. Bitemporal subscenes containing $L = 600 \times 240$ pixels with $M = 202$ after noise elimination are used in the experiment. The false color images of the bitemporal dataset and the ground-truth change map are illustrated in Fig. 12. In Fig. 12(c), the white pixels, gray pixels, and black pixels in the ground-truth change map represent unchanged pixels, changed pixels, and unknown pixels, respectively. The experiment settings follow those presented in Section IV-A with 100 set as the regularization strength. The result of the CD methods conducted on the Bay Area dataset is shown in Table I, and the qualitative results are shown in Fig. 13, with FN and FP marked in green and red, respectively.

Both CVA and PCA demonstrate their ability to prevent false positives, but they also exhibit a tendency to produce

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false negatives. TDRD and CD-ADMM achieve similar performance with the Yancheng dataset with high values of the recall indicator, which means they can identify most of the positive samples. Although GETNET achieves relatively good performance, especially on the precision indicator, aligned to the performance in the Yancheng dataset, the supervised methods suffer from the lack of sufficient training samples to outperform the unsupervised ones. We can find that CODE-HCD yields the same statistical behavior (i.e., the best performance with the highest OA, κ, and precision values), indicating the ability to differentiate the changed and unchanged pixels under the elegant CODE theory [42].

V. Conclusion and Future Work

In the field of HCD, the scarcity of labeled HSI data poses significant challenges to the application of DE-based methods in real-world scenarios. On the other hand, the unsupervised CO algorithms often need mathematically sophisticated regularizers to address the ill-posed problem nature of HCD. In light of these challenges, we propose the CODE-HCD algorithm that blends the advantages of CO and DE by utilizing the recently developed CODE small-data learning theory, which has been adopted to effectively solve various hyperspectral inverse imaging problems. In the proposed CODE-HCD, a convex relaxation domain, together with a convex data-fitting term exploiting the rich spectral information of HSIs, is customized for the HCD problem. In addition, GNN is designed to learn deep features of the HSIs on both pixel level and superpixel level by combining GAT and CNN. Finally, a CODE regularizer with a simple mathematical form of $\boldsymbol{Q}$-quadratic norm is proposed to extract useful information from a roughly estimated solution obtained using a GACDN. Experimental results demonstrate the superiority of the proposed CODE-HCD method under practical semi-supervised conditions of very low sampling/labeling rates. As for future work, though we have successfully achieved significantly lower sampling rates, the semi-supervised method is not fast enough for real-time application. Also, we have been thinking about the possibility of extending the SAM-based data-fitting trick to multiclass CD for a finer understanding of the detected changes.

Appendix
Proof of Theorem 1

Consider the CO problem, i.e.,

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n,\ z\in\mathbb{R}^m} \quad & f_1(\boldsymbol{x}) + f_2(z) \\ \text{s.t. } & \boldsymbol{x} \in \mathcal{S}_1,\ \ z \in \mathcal{S}_2 \\ & z = \boldsymbol{A}\boldsymbol{x} \end{aligned} \tag{24}$$

where $f_1 : \mathbb{R}^n \mapsto \mathbb{R}$ and $f_2 : \mathbb{R}^m \mapsto \mathbb{R}$ are convex functions, $\boldsymbol{A}$ is an $m \times n$ matrix, and $\mathcal{S}_1 \subseteq \mathbb{R}^n$ and $\mathcal{S}_2 \subseteq \mathbb{R}^m$ are nonempty convex sets [18]. The augmented Lagrangian of (24) can be derived as $\mathcal{L}(\boldsymbol{x}, z, \boldsymbol{d}) = f_1(\boldsymbol{x}) + f_2(z) + (c/2)\|\boldsymbol{A}\boldsymbol{x} - z - \boldsymbol{d}\|_2^2$, where $c > 0$ is the penalty parameter. By iteratively updating the primal variable $(\boldsymbol{x}, z)$ and the scaled dual variable $\boldsymbol{d}$, the global optimal solution can be yielded through the ADMM framework, which is summarized in Algorithm 2.

Algorithm 2 General ADMM Algorithm

1: **Given** $c > 0$. Initialize $q := 0$, $\boldsymbol{d}^q$, and $\boldsymbol{x}^q$.
2: **repeat**
3: update $z^{q+1} := \arg\min_{z\in\mathcal{S}_2} \mathcal{L}(\boldsymbol{x}^q, z, \boldsymbol{d}^q)$;
4: update $\boldsymbol{x}^{q+1} := \arg\min_{\boldsymbol{x}\in\mathcal{S}_1} \mathcal{L}(\boldsymbol{x}, z^{q+1}, \boldsymbol{d}^q)$;
5: update $\boldsymbol{d}^{q+1} := \boldsymbol{d}^q - \left(\boldsymbol{A}\boldsymbol{x}^{q+1} - z^{q+1}\right)$;
6: $q := q + 1$;
7: **until** the predefined stopping criterion is met.

As stated in [18, Lemma 9.2], as long as $\boldsymbol{A}^T\boldsymbol{A}$ is invertible, every limit point of the sequence $\{(\boldsymbol{x}^q, z^q)\}$ generated by Algorithm 2 is an optimal solution of problem (24) for any constant positive value of c. A comparison of (21) and (24) reveals that the identity matrix $\boldsymbol{I}_L$ in (21) corresponds to matrix $\boldsymbol{A}$ in (24). In addition, as $\boldsymbol{I}_L^T\boldsymbol{I}_L = \boldsymbol{I}_L$ is also invertible, it follows that every limit point of the sequence $\{\boldsymbol{c}^q\}$ generated by Algorithm 1 is an optimal solution of problem (21). Thus, the statement of Theorem 1 has been proven. ■

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